

Bohuslav Ihorovych Onyshchuk, 10th-grade student at the Regional Scientific Lyceum in Rivne

Research advisor: Volodymyr Andriyovych Syaskyi, Ph.D. in Engineering, Associate Professor, Associate Professor in the Department of Information Technology and Modeling at Rivne State Humanitarian University

Forecasting the Behavior of Time Series Using Machine Learning Methods Based on Kohonen Neural Networks

The aim of this research is to develop an effective approach to clustering and forecasting prices in the financial market using Kohonen neural networks. The ideal outcome is the creation of a model capable of predicting market trends with high accuracy, which will facilitate the making of informed investment decisions.

The object of the study is the forecasting of the behavior of complex systems described by time series.

The subject of the study is machine learning methods using Kohonen networks for clustering and forecasting time series, using the cryptocurrency segment of financial markets as an example.

A time series is a set of measurements of a variable taken over time. In this research paper, the time series consists of Bitcoin price quotes over a four-year period.

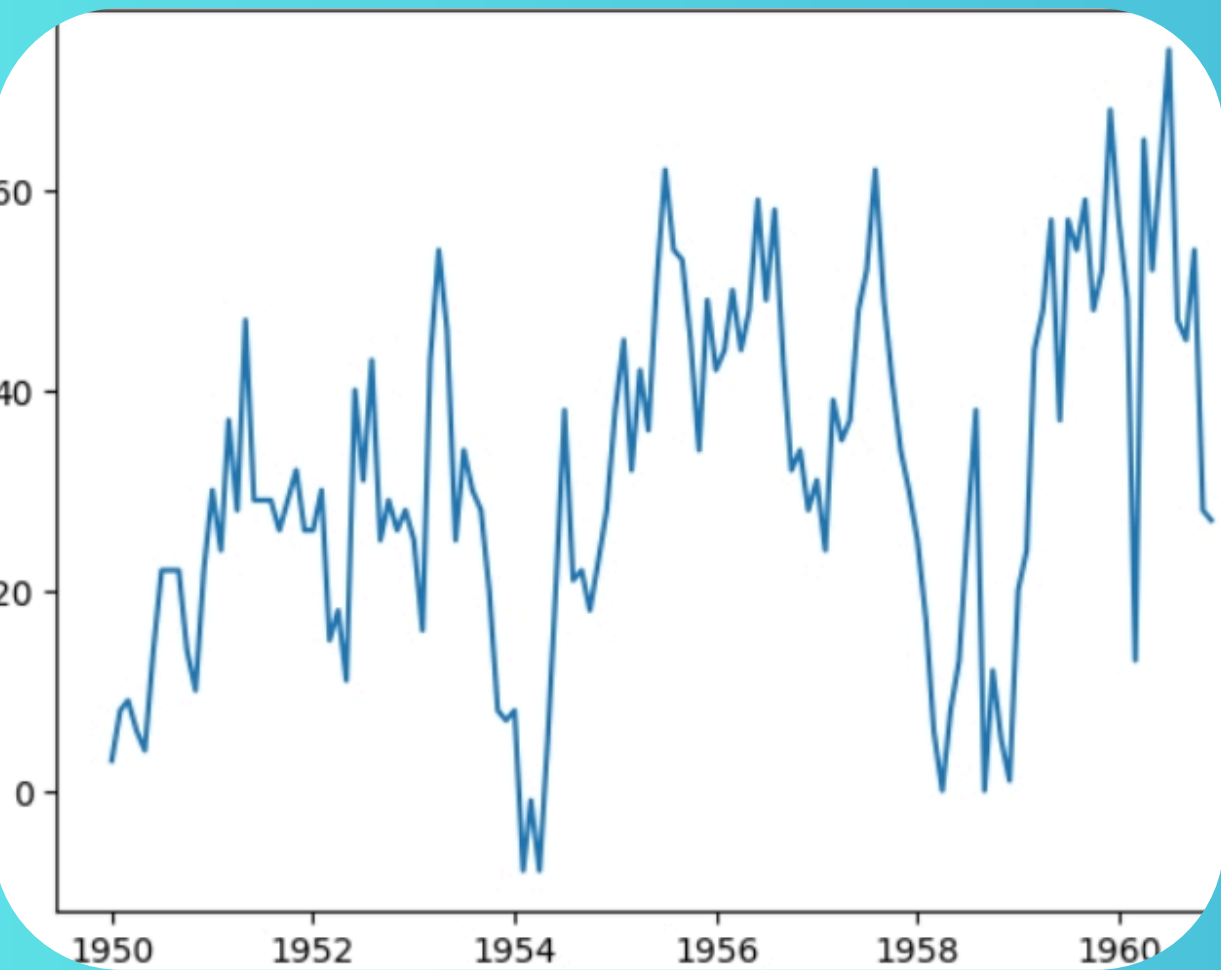


Fig. 1. Example of a time series

The study is based on machine learning-driven data analysis, which enables the identification of hidden patterns in time series. To predict typical sequences of events (typical behavioral patterns in a time series), a training set is formed based on normalized three-week segments, with a step size of 1 week. For clustering, a modified Kohonen network is used (quantization of the training vector, operating on a “winner-takes-all” principle). This tool allows for the clustering of all segments and the identification of typical event sequences, optimizing recognition and forecasting.

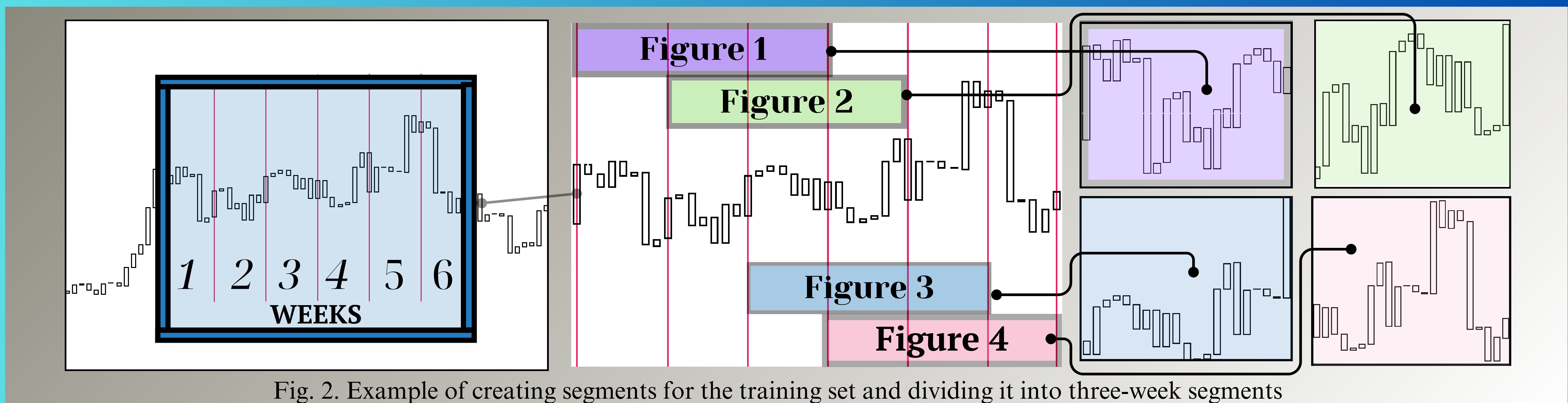


Fig. 2. Example of creating segments for the training set and dividing it into three-week segments

Normalization is used to prepare data for analysis. This process is essential for maintaining the comparability of characteristics across different segments of time series. Normalization helps highlight key changes and ensures that price dynamics are taken into account, which is necessary for subsequent clustering and the development of forecasting models.

Time series forecasting is performed by identifying a partially known segment within the clusters of a Kohonen network. First, a two-week segment from the test dataset is analyzed. This segment is compared with the weight vectors to determine the most similar cluster. The forecast is based on the values of the selected weight vector from the last week. This allows us to assume that the series will continue to follow the pattern characteristic of the cluster.

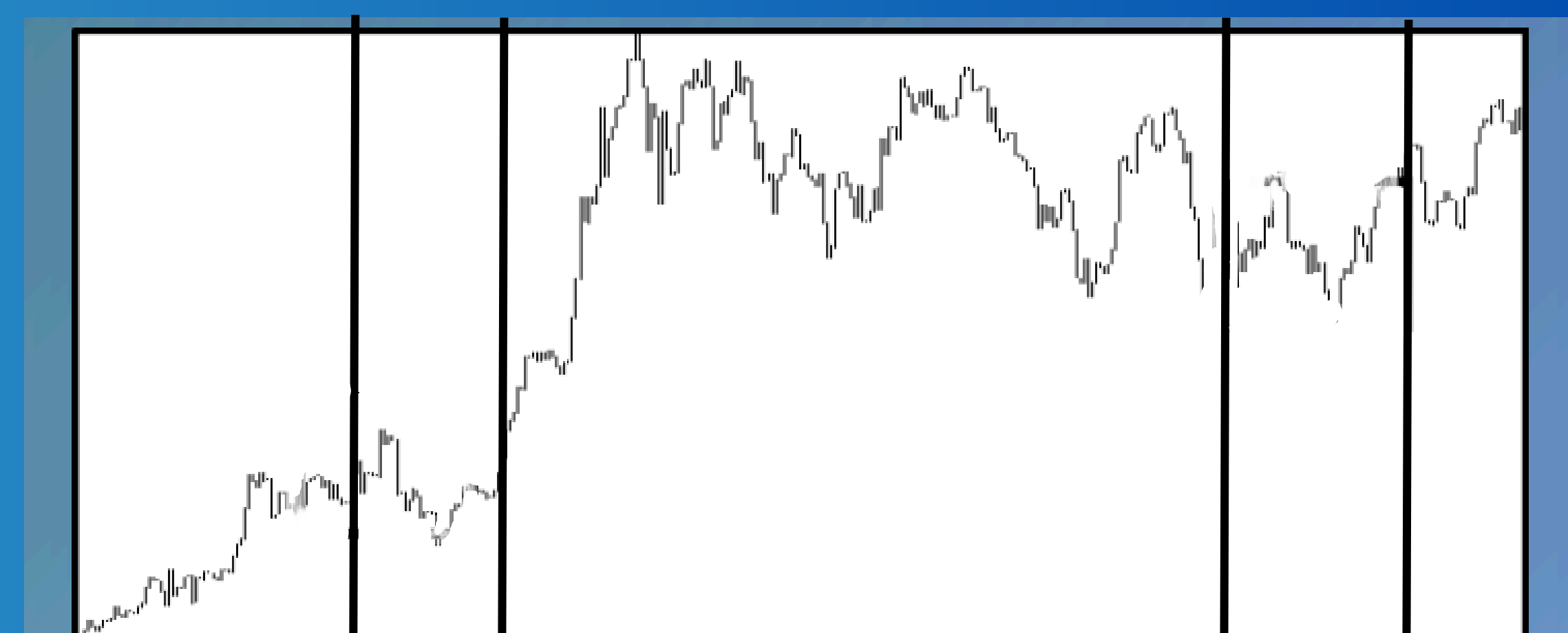


Fig. 3 Similar segments of the time series prior to data normalization



Fig. 4 Similar segments of the time series after normalization

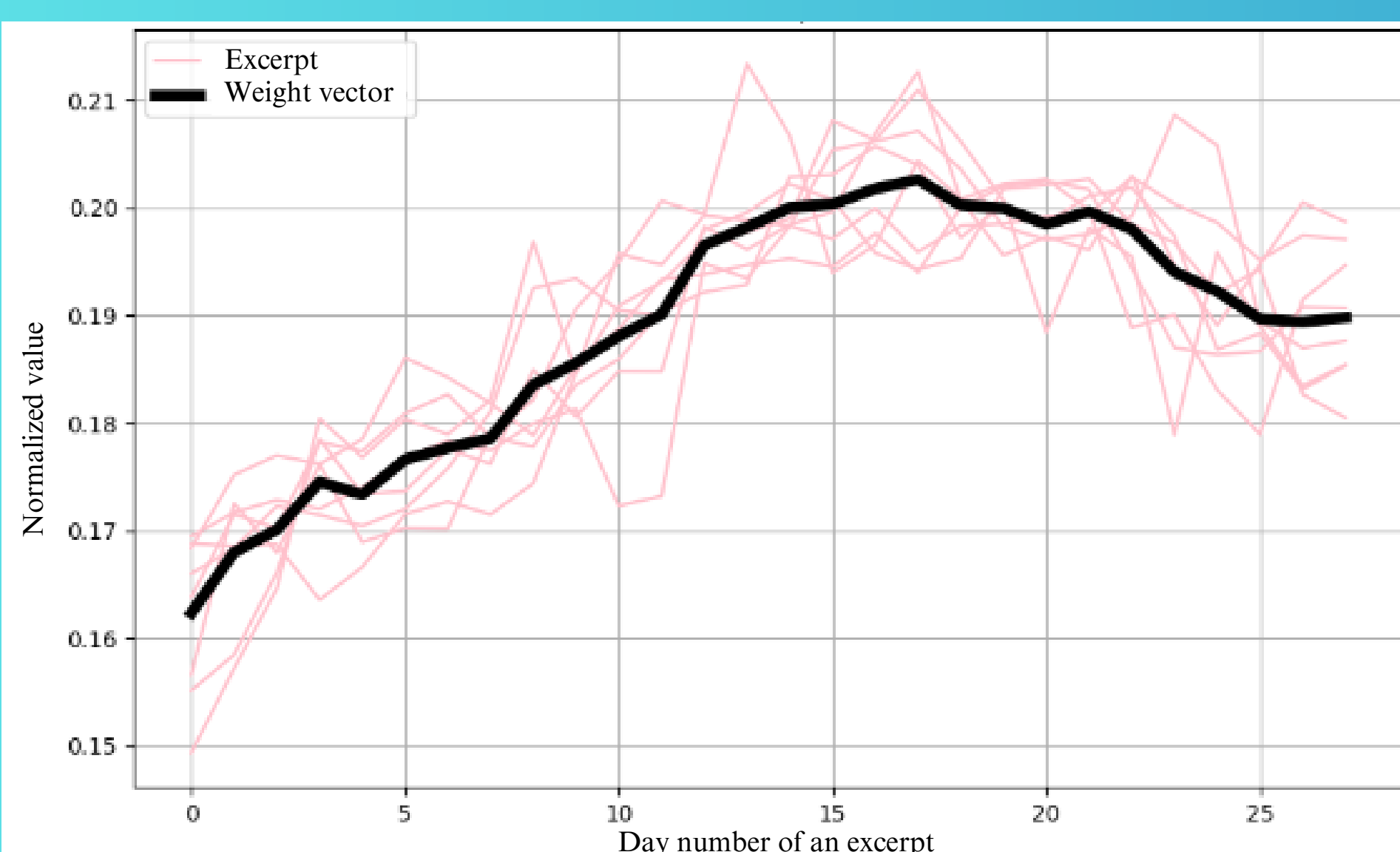


Fig. 5. Example of a formed four-week cluster

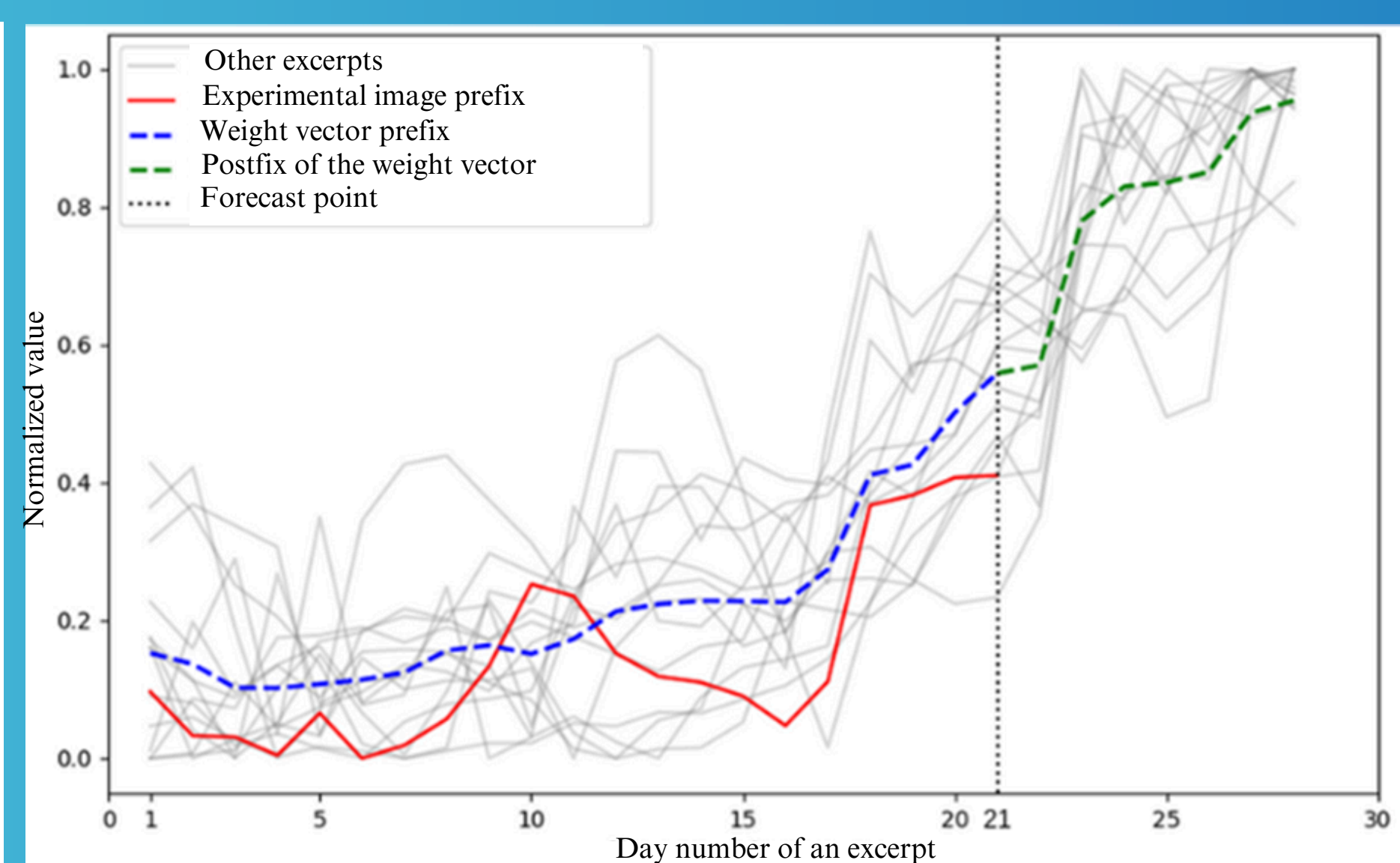


Fig. 6. Example of a generated predictive cluster

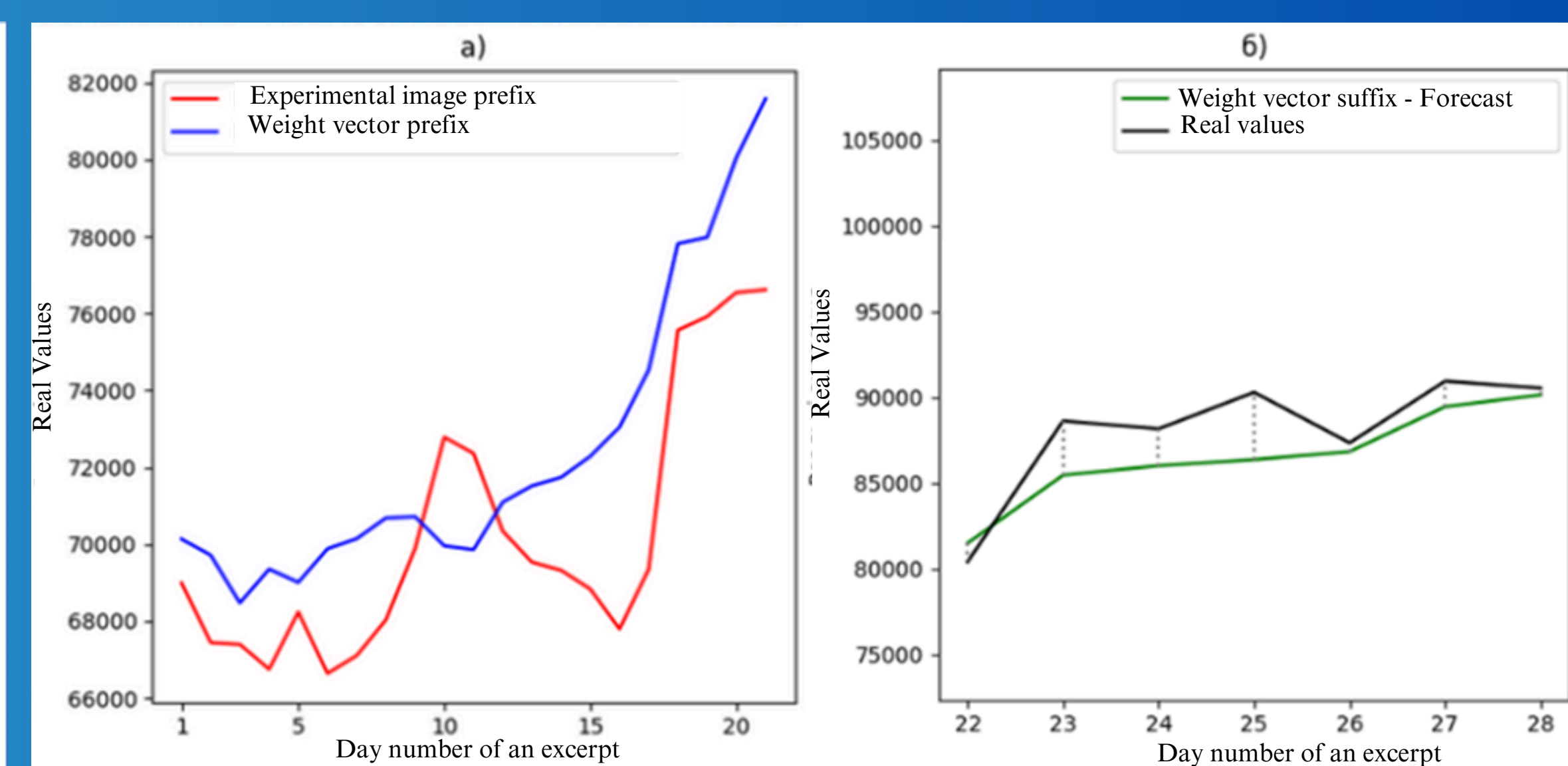


Fig. 7. Example of forecasting real data for a segment

Conclusions: The proposed approach, which combines clustering and Kohonen neural networks, has demonstrated acceptable forecasting accuracy and adaptability to market changes. The developed program is versatile and can be applied not only to financial data but also to the analysis of any cyclical phenomena—in particular, climate events such as El Niño/La Niña or the spread of pandemic diseases. The method has potential for further refinement and broader application in various fields of intelligent data analysis.